A-A-51143
29 March 1985

COMMERCIAL ITEM DESCRIPTION

CUTTER BIT, TOOL HOLDER (CARBIDE TIP)

The General Services Administration has authorized the use of this commercial item description in preference to Federal specification 00-C-00845.

This description covers cutter bits with brazed sintered carbide tips that are supported by a tool holder and used on metal cutting machines for taking roughing to finishing cuts on cast iron, steel, non-ferrous, and non-metallic materials.

Salient characteristics.

Classification. This description covers cutter bits of the following styles, sizes, and classes. The style, size, and class cutter bit required for a specific procurement shall be as selected from tables I, II, and III and salient characteristics as listed in this description.

Style AR - 0 degree side cutting edge angle, right hand.
Style AL - 0 degree side cutting edge angle, left hand.
Style BR - 15 degree side cutting edge angle, right hand.
Style BL - 15 degree side cutting edge angle, left hand.

Size 4 - 1/4 inch wide x 1/4 inch high.
Size 5 - 5/16 inch wide x 5/16 inch high.
Size 6 - 3/8 inch wide x 3/8 inch high.
Size 7 - 7/16 inch wide x 7/16 inch high.
Size 8 - 1/2 inch wide x 1/2 inch high.
Size 10 - 5/8 inch wide x 5/8 inch high.
Size 12 - 3/4 inch wide x 3/4 inch high.
Size 16 - 1 inch wide x 1 inch high.

Class. The class of carbide tool specified shall be as selected from table I according to typical use.

Design. The cutter bit style and size shall conform to the requirements of ANSI B94.5 "Carbide Blanks and Single-Point Brazed Tools."

Dimensions. The dimensions of the cutter bits shall conform to tables II and III for the style and size required for a specific procurement. The tolerances for all dimensional data shall be the manufacturers standard as specified on their part drawings.

FSC 3455

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TABLE I. Intended uses, typical machining applications.

Class	Commercial designation grade	*Typical machining operations	*Typical materials	Typical cuts	
				Cutting speed	Tool feed
1	C-1	Heavy duty roughing cuts by turning, milling, shaping and planing under favorable machining conditions	Low hardness gray cast iron, metal that generate short chips, non-ferrous metals, low tensile strength steel, stainless steel, non-metallic materials	Low	Coarse
2	C-2	Semi-roughing and general purpose machining by turning, milling, shaping, planing, and boring	Gray cast irons having hardnesses up to Rockwell C 20, non-ferrous metals, titanium metals, high temperature steel alloys, monel, bronze alloys, brasses	Medium	Medium
3	C-3	Semi-finishing and finishing cuts by turning, milling, drilling, boring, and broaching	Gray cast irons over Rockwell C 20 hardness, malleable iron and nodular iron that generates short chips, hardened steel, silicon aluminum alloys, titanium metals, plastics, hard rubber, fiberglass	Medium to high	Medium to fine
4	C-4	Finishing cuts, precise dimension control, and best surface finishes by turning, boring, and milling	Very hard gray iron castings and chilled castings harder than Rockwell C 60, high silicon aluminum alloys, hardened steel, highly abrasive plastics, phenolic laminates, ceramics	High	Fine
5	C-5	Heavy duty roughing cuts by turning, milling, shaping, and planing under unfavorable machining conditions	Carbon and alloy steel, steel castings with cored sections, cavities, sand inclusions, malleable iron that generates relatively long chips	Medium to low	Medium to coarse
6	C-6	Semi-roughing and general purpose machining by turning, milling, shaping and planing	Carbon and alloy steels, steel castings, malleable iron that generates relatively long chips	Medium	Medium
7	C-7	Semi-finishing and finishing cuts by turning, milling, threading and boring	Carbon and alloy steels, steel castings, stainless steels, malleable and nodular irons that generate relatively large chips	Medium to high	Medium to fine
8	C-8	Fine finishing cuts, fine surface finishes, and precise dimension control by turning and boring. Precise threading by high speed, single point threading machines	Carbon and alloy steels, steel castings	High	Fine

*Note: "Typical operations" and "Typical materials" are listed in the approximate order of decreasing frequency of usage, i.e. most frequent applications are listed first followed by less frequent applications.

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TABLE II. Style AR and AL cutter bit size and dimensions.

Tool dimensions, inches						Tip dimensions, inches			
Style-Size		Width	Height	Length	Nose radius	Tip overhang	Thick	Width	Length
AR-4	AL-4	1/4	1/4	2	1/64	1/32	3/32	3/16	5/16
AR-5	AL-5	5/16	5/16	2-1/4	1/64	1/32	3/32	1/4	1/2
AR-6	AL-6	3/8	3/8	2-1/2	1/64	1/32	3/32	1/4	1/2
AR-7	AL-7	7/16	7/16	3	1/32	1/32	3/32	1/4	1/2
AR-8	AL-8	1/2	1/2	3-1/2	1/32	1/32	1/8	5/16	5/8
AR-10	AL-10	5/8	5/8	4	1/32	1/32	5/32	3/8	3/4
AR-12	AL-12	3/4	3/4	4-1/2	1/32	1/32	3/16	7/16	13/16
AR-16	AL-16	1	1	7	1/32	1/16	1/4	9/16	1
AR-16A	AL-16A	1	1	6	1/32	1/16	1/4	9/16	1

TABLE III. Style BR and BL cutter bit size and dimensions.

Tool dimensions, inches						Tip dimensions, inches			
Style-Size		Width	Height	Length	Nose radius	Tip overhang	Thick	Width	Length
BR-4	BL-4	1/4	1/4	2	1/64	1/32	1/16	5/32	1/4
BR-5	BL-5	5/16	5/16	2-1/4	1/64	1/32	3/32	3/16	5/16
BR-6	BL-6	3/8	3/8	2-1/2	1/64	1/32	3/32	1/4	1/2
BR-7	BL-7	7/16	7/16	3	1/32	1/32	3/32	1/4	1/2
BR-8	BL-8	1/2	1/2	3-1/2	1/32	1/32	1/8	5/16	5/8
BR-10	BL-10	5/8	5/8	4	1/32	1/32	5/32	3/8	3/4
BR-12	BL-12	3/4	3/4	4-1/2	1/32	1/32	3/16	7/16	13/16
BR-16	BL-16	1	1	7	1/32	1/16	1/4	9/16	1
BR-16A	BL-16A	1	1	6	1/32	1/16	1/4	9/16	1

Material. The tool body shall be steel with a brazed sintered carbide cutting tip.

Measurement system. All dimensions in this commercial item description are given in the English system of measurement. These dimensions can be converted to Metric system equivalents by the use of FED-STD-376.

Marking. The body of the tool shall be permanently marked with the manufacturer's name or trade mark.

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Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the commercial item description where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

Acceptance inspection. The supplier shall randomly select the number of samples specified in table IV that is applicable to the quantity of items being purchased. Each sample shall be visually inspected, measured, and tested by the supplier to determine conformance with the requirements specified herein. Any sample failing to meet these requirements shall be considered a defective item. If the number of defective items equals, or is less than the quantity specified under "Acceptance, number defective" column of table IV, the supplier shall replace the defective samples with acceptable items and the entire lot shall be considered acceptable for shipment. If the number of defective samples equals, or exceeds, the quantity specified in the "Rejection, number defective" column of table IV, the entire lot shall be considered defective and unsuitable for acceptance.

TABLE IV. Sampling plan.

Quantity of items being purchased	Sample size	Acceptance, number defective	Rejection, number defective
2 - 8	2	0	1
9 - 15	3	0	1
16 - 25	5	0	1
26 - 50	8	1	2
51 - 90	13	1	2
91 - 150	20	2	3
151 - 280	32	3	4
281 - 500	50	5	6

Regulatory requirements. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable.

Preservation, packaging, labeling, and marking. Unless otherwise specified in the contract or order, preservation, packaging, packing, labeling, and marking shall be in accordance with ASTM D 3951. When specified in the contract or order, preservation, packaging, packing, labeling, and marking shall be in accordance with Federal specification PPP-T-1150, level as specified.

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Notes. ANSI B94.5 "Carbide Blanks and Single-Point Brazed Tools" is available from the American National Standards Institute, 1430 Broadway, New York, NY 10018.

ASTM D 3951 "Packaging, Commercial" is available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

FED-STD-376 "Preferred Metric Units for General Use by the Federal Government" and Federal Specification PPP-T-1150 "Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery, Packaging and Packing Of" should be obtained from the contracting activity or as directed by the contracting officer.

Purchaser should specify the following:

a. Title, number, and date of this commercial item description.

b. Quantity, style, size, and class of cutter bit.

c. Special packaging in accordance with PPP-T-1150 and level, if required.

The carbide class shall be based on the material being machined.

The style, size, and tool numbers in this description correspond to the style, size, and tool numbers in Federal Specification 00-C-00845.

MILITARY INTERESTS:

Custodians

Army - AL
Navy - SH
Air Force - 99

Review Activities

Air Force - 84
DLA - GS

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

DLA - IP

Project 3455-0094

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-51143	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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